

Linear Algebra-Lab01: Arithmetic Refresher

1.1. Basic exercises

For all underneath exercises do at least one pen and paper attempt in your own workbook, before checking your results in GeoGebra. In case of mismatches between your handwritten results and GeoGebra's output, do not hesitate to seek assistance by your Lab attendant, as you are owner of your own learning.

1.1.1. Arithmetics

Exercise 1: Simplify the following expressions (GeoGebra/View/CAS only)

- | | |
|--------------------|--------------------------|
| 1) $a^5 \cdot a^5$ | 5) $a^{10} \cdot a^{-3}$ |
| 2) $a^5 + a^5$ | 6) $\frac{a^8}{a^2}$ |
| 3) $a^5 - a^5$ | 7) $\frac{a^6}{a^{-2}}$ |
| 4) $(a^5)^3$ | 8) $(2a^{-3})^{-2}$ |

Exercise 2: Tick every correct answer (GeoGebra/View/CAS, Simplify)

1)	$-(a+b) =$	$-a-b$	$a-b$	$-a+b$	$a+b$
2)	$-\frac{a+b}{c} =$	$\frac{a+b}{-c}$	$\frac{-a+b}{c}$	$\frac{-a-b}{c}$	$\frac{-a-b}{-c}$
3)	$\frac{c}{d} + \frac{3}{4} =$	$\frac{c+3}{d+4}$	$\frac{c+3}{4d}$	$\frac{4c+3d}{4d}$	$\frac{4c+3d}{d+4}$
4)	$\frac{c}{d} \cdot \frac{3}{4} =$	$\frac{c+3}{d+4}$	$\frac{4d}{3c}$	$\frac{3c}{4d}$	$\frac{12c}{4d}$
5)	$\left(\frac{-1}{7}\right)^{-1} =$	$\frac{1}{7}$	-7	$\frac{-1}{7}$	7
6)	$a^2 \cdot a^n \cdot a^m =$	a^{2nm}	a^{2+n+m}	a^{2+nm}	a^{2n+m}
7)	$(-2a^2b^3)^3 =$	$-6a^6b^9$	$-6a^2b^6$	$-8a^6b^9$	$-8a^8b^{27}$
8)	$\frac{c^6d^3}{c^2d} =$	c^3d^3	c^4d^3	c^4d^2	c^3d^2

1.1.2. Polynomial algebra

Exercise 3: Simplify as much as possible (GeoGebra/View/CAS, Simplify)

In GeoGebra the product $x y$ is different from the string xy – **be aware!**

In GeoGebra the division operator to insert is / (slash) instead of : (colon).

- | | | |
|-------------------------|--------------------------|------------------------------------|
| 1) $-4x^2y^2 + (2xy)^2$ | 4) $(x^2y)^3(-x^6y^3)$ | 7) $x(x-1) + (2x+1)(x-3)$ |
| 2) $-4x^2y^2 + (2xy)^2$ | 5) $(x^2y)^3(-x^6y)^3$ | 8) $6x(x+5) - 8x(x^2 - 4x + 3)$ |
| 3) $-4x^2y^2 : (2xy)^2$ | 6) $(x^2y)^3 - (x^6y^3)$ | 9) $(3x-2)(x-4) - 3x^2 + 7x(2x-1)$ |

1.1.3. Equation solving

Exercise 4: Solve only the *linear* equations in the unknown x from underneath list (GeoGebra/View/CAS, Solve(button))

- | | |
|--|---|
| 1) $3 - (2x - 1) = 3(x - 2)$ | 5) $\frac{4}{5}x - \frac{1}{3} = \frac{5}{6}x$ |
| 2) $\frac{x+6}{3} - \frac{5x+3}{12} = 2$ | 6) $-3(x-5) = 4\left(-x + \frac{1}{2}\right) - (7-10x)$ |
| 3) $5x^2 + 3 = 4(x+1)$ | 7) $8x + 1 = 6x + 7$ |
| 4) $3x^2 + 2x = 1$ | 8) $2x - 1 = 4x - 5$ |

Exercise 5: Solve these quadratic equations for their unknown, applying the abc-formulas aka discriminant (GeoGebra/View/CAS, Solve(button), alpha(button))

- | | |
|------------------------------|--------------------------------------|
| 1) $6\delta^2 + 5\delta = 0$ | 4) $4x^2 + 5x + 1 = 0$ |
| 2) $4x^2 - 25 = 0$ | 5) $(t+1)^2 + (t+2)^2 + (t+3)^2 = 2$ |
| 3) $2t^2 - t - 6 = 0$ | 6) $t(t+1) = 6$ |

1.1.4. Trigonometric fundamentals: angular units

Exercise 6: Convert accordingly to radians, or to degrees and simplify the result.

Mind to click the $^\circ$ symbol, or type Alt+o for degree input (GeoGebra/View/CAS)

- | | |
|-----------------|----------------------|
| 1) -30° | 6) $\frac{1}{2}\pi$ |
| 2) 450° | 7) $\frac{-4}{3}\pi$ |
| 3) -160° | 8) 12π |
| 4) 60° | 9) $\frac{1}{6}\pi$ |
| 5) 15° | 10) $\frac{3}{6}\pi$ |

1.2. Bridging exercises

1.2.1. Exploring polynomials

Exercise 7: Calculate the polynomial's return value given its input value (GeoGebra/View/CAS, Substitute(button))

a) $5x^3 - 7x^2 + 8x - 3$ in $x = 2$

e) $11 - x^2 + 2x^3$ in $x = -2$

b) $2x^3 - x^2 - 7x + 2$ in $x = -1$

f) $x^4 + x^3 - 10x - 6$ in $x = 5$

c) $x^3 + 5x - 4$ in $x = 3$

g) $2x^4 - 2x^2 + 15x$ in $x = -6$

d) $6 - x - 2x^2 - x^3$ in $x = -1$

h) $x^4 + x^3 - x^2 + 1$ in $x = \frac{2}{3}$

The first screenshot shows the GeoGebra CAS view with the 'Substitute' button highlighted. A tooltip for the 'Substitute' button reads: 'Substitute Enter replacement for part of expression'. The input field contains the polynomial $5x^3 - 7x^2 + 8x - 3$.

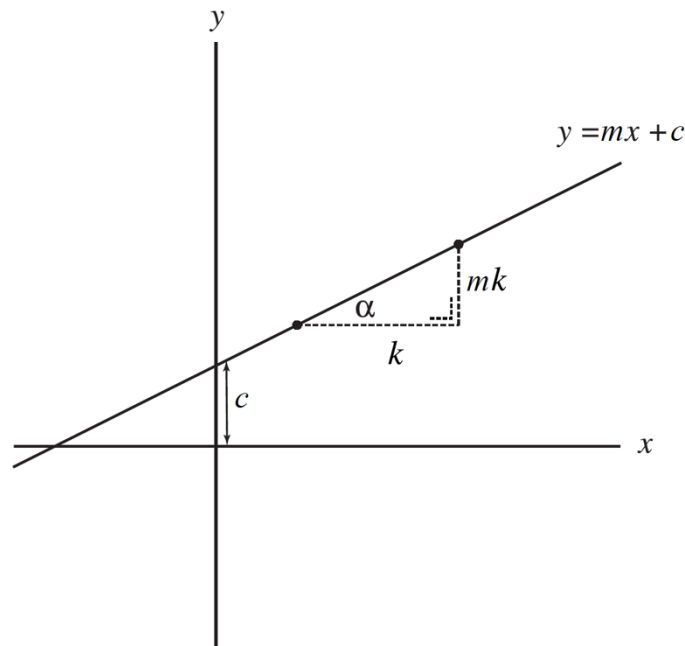
The second screenshot shows the 'Substitute - Row 2' dialog box. It has two columns: 'Old expression' and 'New expression'. The 'Old expression' column contains 'x' and the 'New expression' column contains '2'.

1.2.2. Visualize the solution sets of equations

Exercise 8: Linear expressions in x simplified to the standard form $mx+c$ draw straight lines.

The final coefficient m of variable x denotes **the slope** of this line: for each k steps (to the right) the line rises mk steps up ($m>0$) or down ($m<0$).

The **intercept** c simply shows on which height this line is intersecting the y -axis.



Graph (pen and paper) the **simplified** left hand and right hand **sides** of only the *linear* equations over a horizontal running x-axis, as their intersection shows the solution set of their equation. (GeoGebra/View/CAS + /Algebra + /Graphics)

1) $3 - (2x - 1) = 3(x - 2)$

5) $\frac{4}{5}x - \frac{1}{3} = \frac{5}{6}x$

2) $\frac{x+6}{3} - \frac{5x+3}{12} = 2$

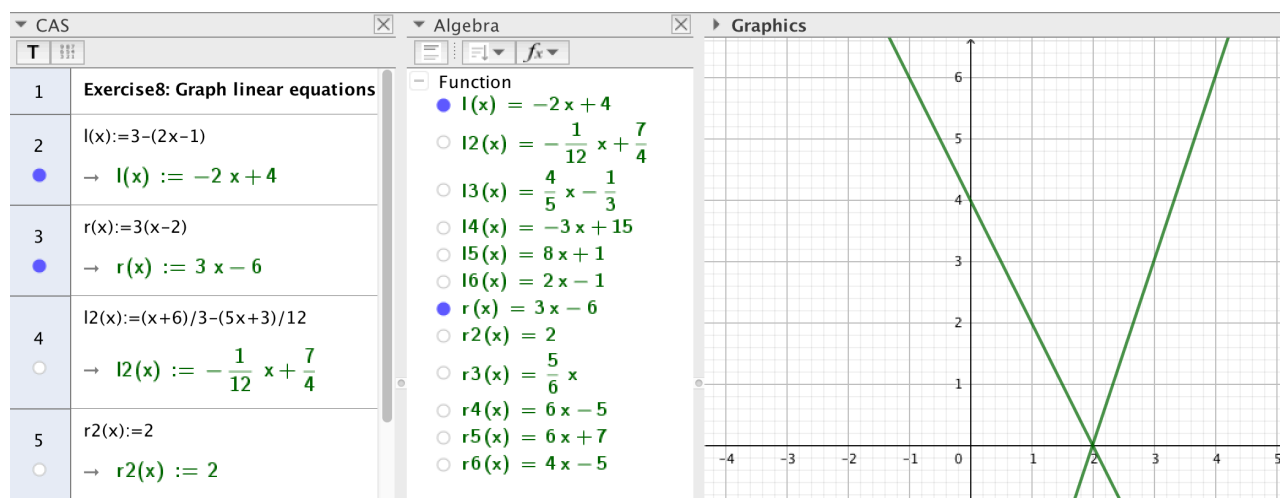
6) $-3(x - 5) = 4\left(-x + \frac{1}{2}\right) - (7 - 10x)$

3) $5x^2 + 3 = 4(x + 1)$

7) $8x + 1 = 6x + 7$

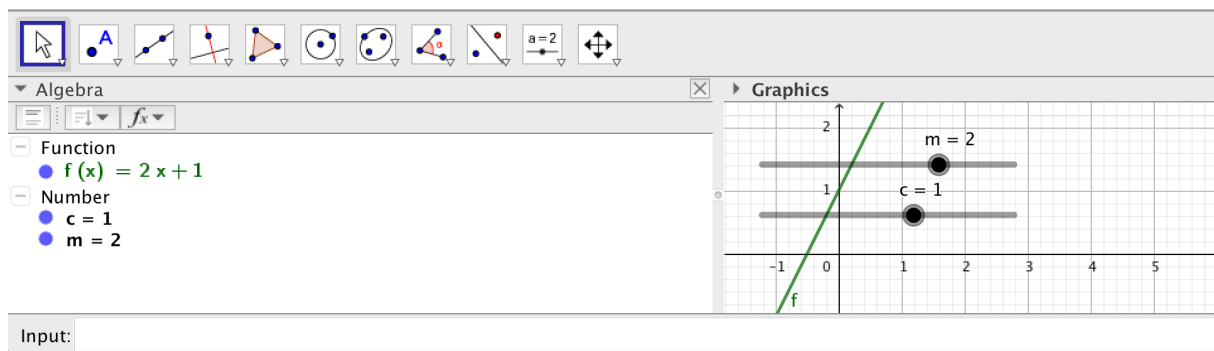
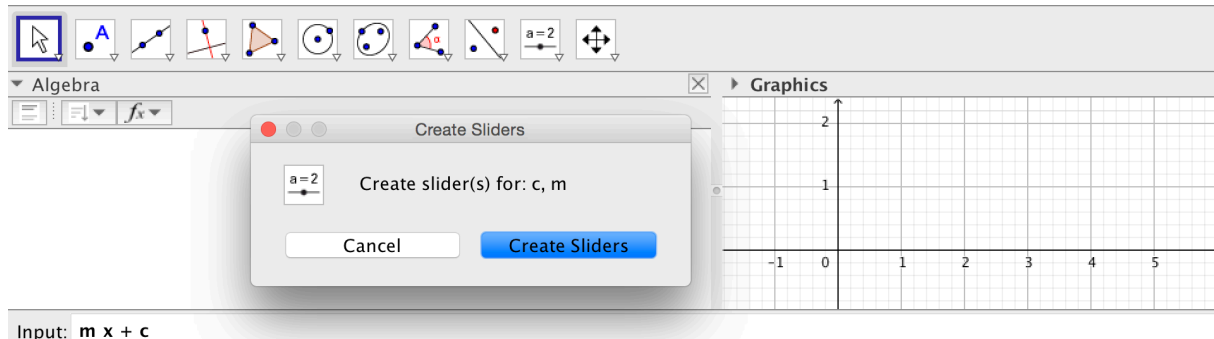
4) $3x^2 + 2x = 1$

8) $2x - 1 = 4x - 5$



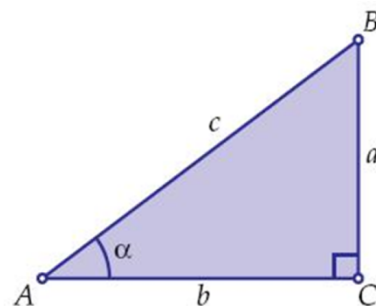
Exercise 9: As linear expressions in x simplified to the standard form $mx+c$ draw straight lines which slope by coefficient m and intercept by the constant c , explore this line(graph) in GeoGebra by varying both m and c by using sliders.

(GeoGebra/View/Algebra + /Graphics + /Input Bar)



1.2.3. Trigonometric fundamentals: right triangles

Exercise 10: Solve each of the six underneath right triangles given, only for its side a whilst applying the shortest strategy. An expression suffices to answer a .



a) $\alpha = 30^\circ, c = 3$

b) $\alpha = 45^\circ, c = 2$

c) $\alpha = 60^\circ, c = 7$

d) $\alpha = 45^\circ, b = 8$

e) $\alpha = 60^\circ, b = 2$

f) $\alpha = 15^\circ, b = 3$

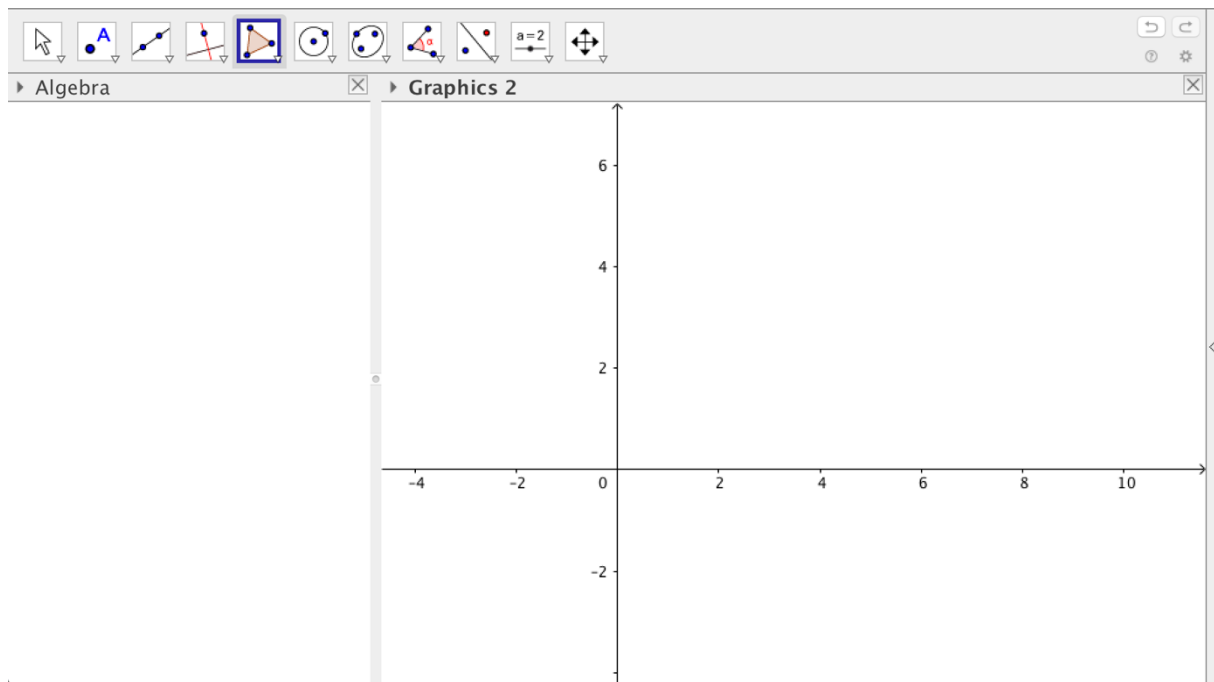
(GeoGebra/View/CAS, Symbolic Evaluation(button))

1	Exercise10: Solve only for a
2	Solve(sin(30°)=a/3) $\rightarrow \left\{ a = \frac{3}{2} \right\}$
3	Solve(sin(45°)=a/2) $\rightarrow \left\{ a = \sqrt{2} \right\}$
4	Solve(sin(60°)=a/7) $\rightarrow \left\{ a = 7 \cdot \frac{\sqrt{3}}{2} \right\}$
5	Solve(tan(45°)=a/8) $\rightarrow \left\{ a = 8 \right\}$
6	Solve(tan(60°)=a/2) $\rightarrow \left\{ a = 2 \sqrt{3} \right\}$
7	Solve(tan(15°)=a/3) $\rightarrow \left\{ a = -3 \sqrt{3} + 6 \right\}$

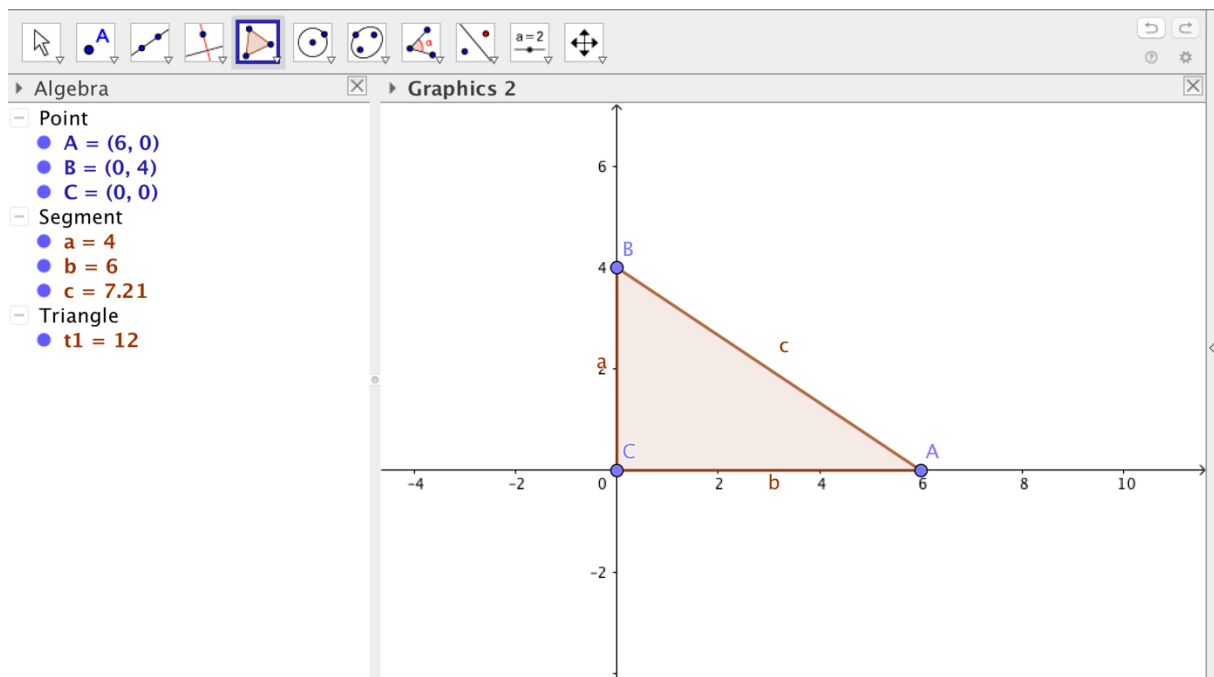
Experience GeoGebra by displaying the previous answers numerically up to 5 significant digits. (GeoGebra/View/CAS, Numeric Evaluation (button))

1	Exercise10: Solve only for a
2	Solve(sin(30°)=a/3) $\approx \left\{ a = 1.5000 \right\}$
3	Solve(sin(45°)=a/2) $\approx \left\{ a = 1.4142 \right\}$
4	Solve(sin(60°)=a/7) $\approx \left\{ a = 6.0622 \right\}$
5	Solve(tan(45°)=a/8) $\approx \left\{ a = 8.0000 \right\}$
6	Solve(tan(60°)=a/2) $\approx \left\{ a = 3.4641 \right\}$
7	Solve(tan(15°)=a/3) $\approx \left\{ a = 0.80385 \right\}$

Exercise 11: As right triangles feature their constant angle of 90° opposite the hypotenuse, we create this polygon in GeoGebra stepwise like underneath. (GeoGebra/View/Algebra + /Graphics2, Polygon(button)) Click its three vertices subsequently in the Graphics2-screen putting the right angle in the origin (0,0).

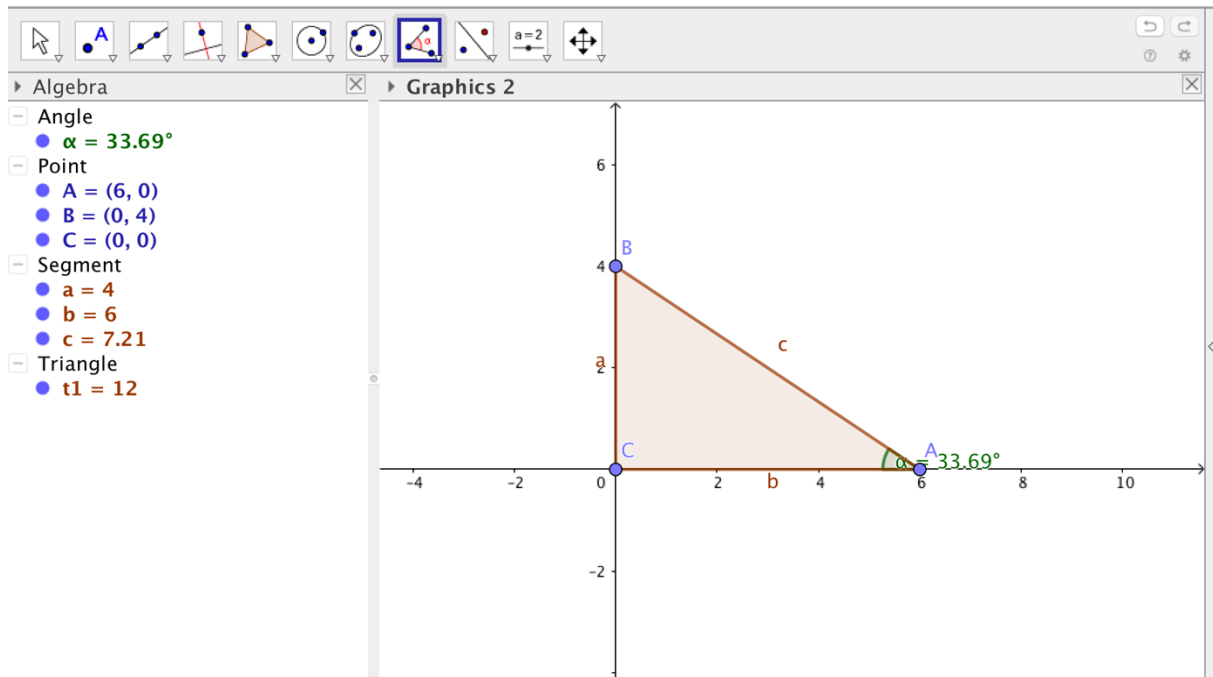


GeoGebra applies the naming convention (side a laying opposite vertex A , etc).



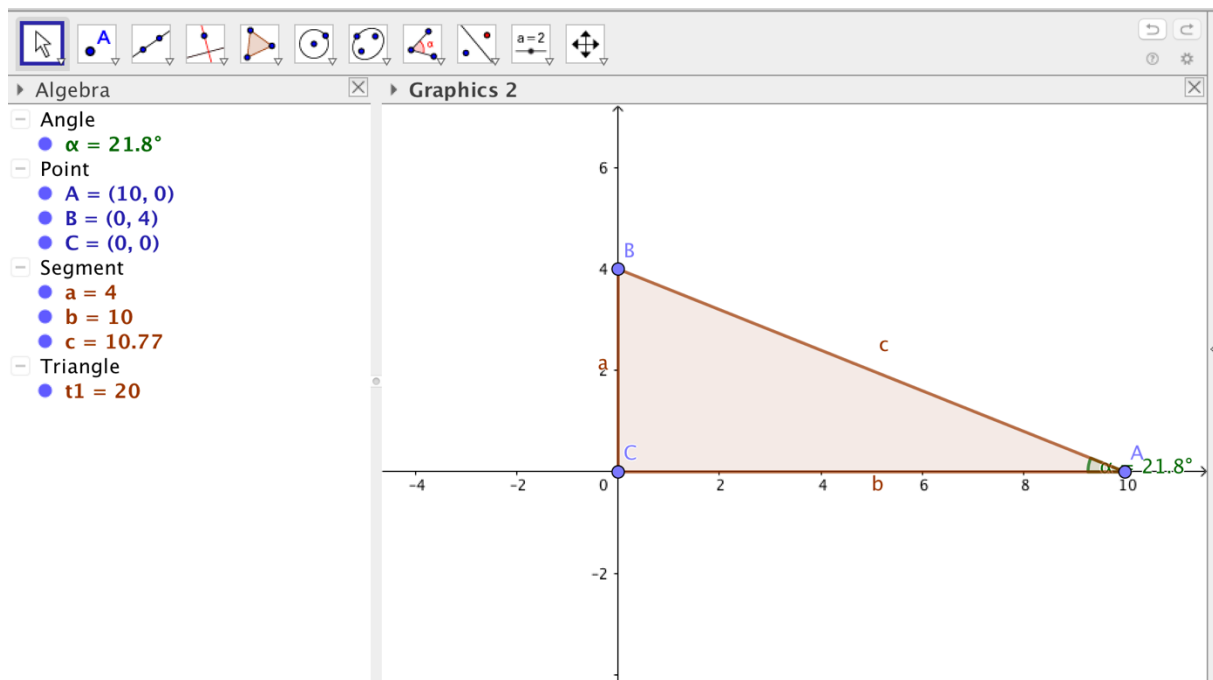
Nextly we overlay in this polygon only **interior angle** α by clicking appropriately three points with the Angle(button) in the Graphics2-screen.

(GeoGebra/View/Algebra + /Graphics2, Angle(button))



Finally we can explore and visualize some cases, for instance of the previous exercise answers by dragging either vertex A horizontally or vertex B vertically using the Move(button) in the Graphics2-screen.

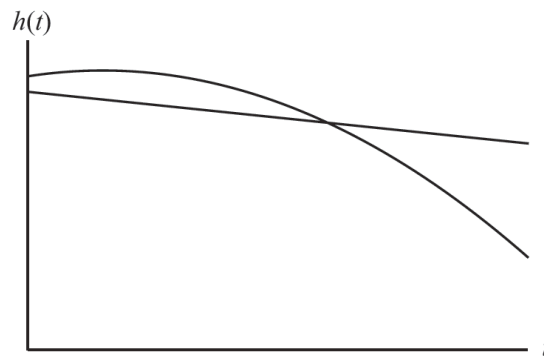
(GeoGebra/View/Algebra + /Graphics2, Move(button))



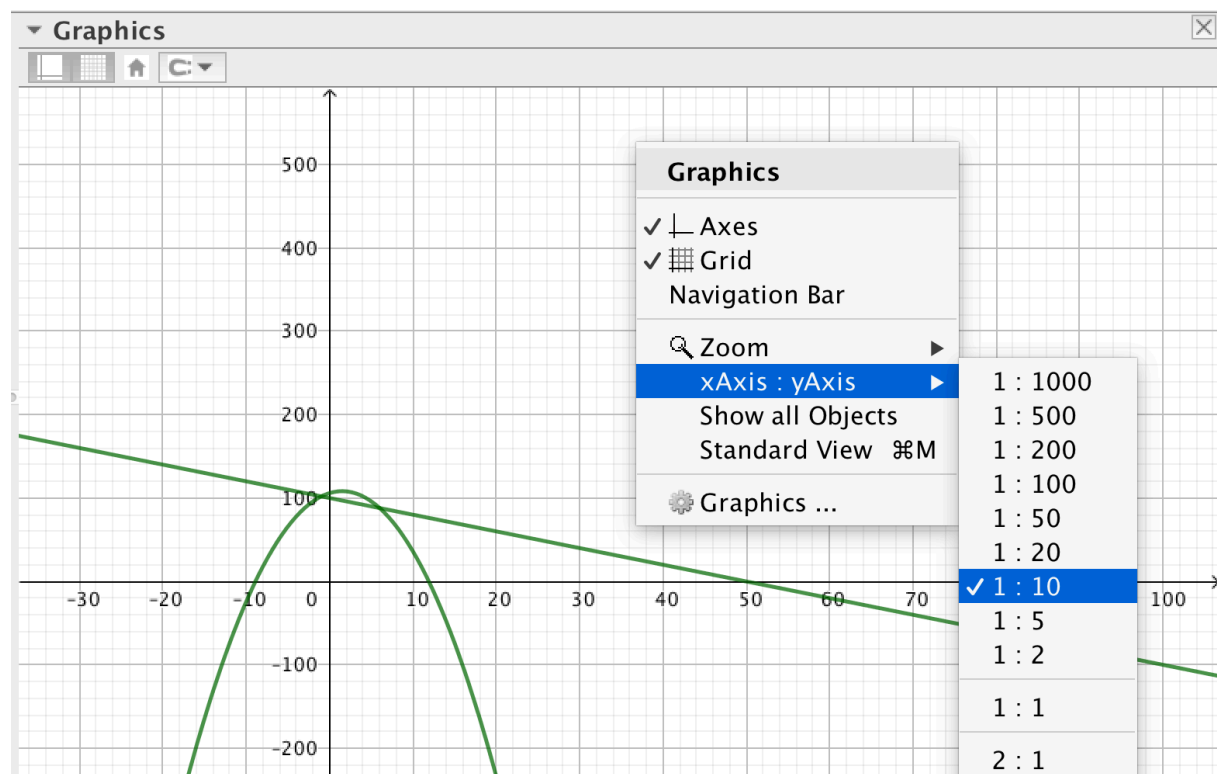
1.3. Contextual practice

1.3.1. Intersection problem

Exercise 12: Underneath graphs the heights h of both the descending straight line $-2t+100$ and the curved line $-t^2+3t+106$ and shows their intersection for a positive t .

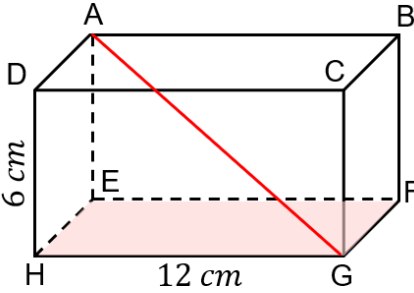
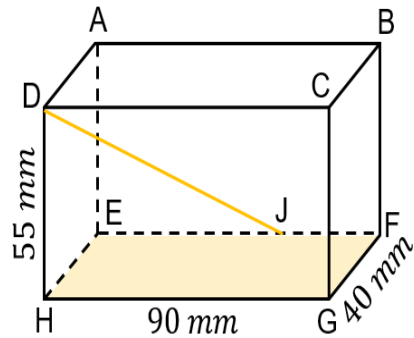
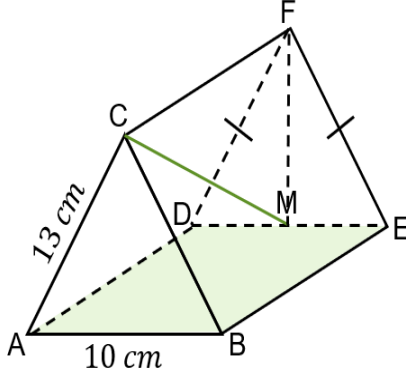
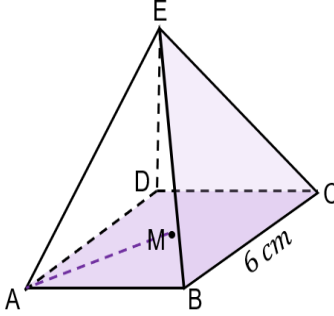


- 1) Solve their equal heights for t by pen & paper (4 decimal places rounding)
- 2) Verify the previous labour by GeoGebra (GeoGebra/View/CAS)
- 3) Determine their corresponding equal height (for the positive solution t)
- 4) Reveal the positive solution t graphically (GeoGebra/View/CAS+Graphics)



1.3.2. Pythagorean problems

Exercise 13 (a) (b) (c) (d): Find the features asked for each of these four underneath **3D solids**. Use pen and paper to reason and GeoGebra to calculate all values rounded up to 4 decimal places. Express angles both in radian and in (decimal) degrees up to 4 decimal places. Answer lengths consequently in the units given (cm, mm, ...).

(a) Cuboid The volume of the cuboid ABCDEFGH equals 360 cm^3.  Find 1) side's length AD 2) diagonal's length AG 3) the angle subtended by diagonal $[AG]$ and the basis plane EFGH	(b) Point J splits the edge $[EF]$ in the ratio $2 : 1$.  Find 1) horizontal length HJ 2) length DJ 3) the angle HDJ (so in point D) 4) the angle subtended by $[DJ]$ and the basis plane EFGH
(c) Prism ABCDEF is a triangular right prism with a total surface area of 660 cm^2. M is the midpoint of $[DE]$.  Find 1) length of its back face altitude MF 2) side's length BE 3) diagonal's length CM 4) the angle subtended by $[CM]$ and the shaded horizontal face ABED	(d) Pyramid The volume of the square-based pyramid ABCDE is 180 cm^3. M is the centroid of the base which lies vertically below E.  Find 1) height ME 2) length AE 3) angle EAM 4) the angle subtended by the isosceles triangular side face BCE and the square basis plane ABCD